

Business Problem

ITSM Service Desk Operations Analytics Dataset

ITSM Ticket, SLA & Support Operations Analytics using Python & PostgreSQL

1. Project Background

This project focuses on an IT Service Management (ITSM) environment where a managed IT services company supports multiple enterprise clients. The organization generates and manages service tickets through a service desk.

Each ticket may involve incidents, service requests, access requests, change requests, or problems.

The purpose of the analysis is to use the available ticket, SLA, company, user and operational KPI data to understand how the service desk is performing and where operational improvements may be required.

2. Business Problem

The main business problem is to get a clear, data-based view of IT service performance.

The project mainly focuses on three areas:

- Workload management
- SLA compliance
- Operational reporting

Workload Management

The service desk handles a large number of tickets across different companies, categories, priorities and support teams.

Management needs to understand where the demand is concentrated and how the workload is distributed.

This information can help identify areas where queue management, staffing, ticket routing or support processes may need attention.

SLA and Service Performance

Tickets are measured against First Response and Resolution SLA targets.

The business needs to understand:

- How often SLAs are met or breached
- Which priorities or support areas have higher breach levels
- How resolution time is related to service performance.

Operational Reporting

The database contains operational KPI and management-summary information.

The analysis should help convert these records into understandable KPIs, trends and business insights that can support management decisions.

3. Main Business Questions

The project aims to answer questions such as:

- How is ticket demand distributed across ticket types, categories, priorities and source channels?
- Which companies, regions and support teams generate or handle the highest ticket volumes?
- How does resolution time vary across ticket priorities, categories and support teams?
- Where are First Response and Resolution SLA breaches concentrated?
- How does the backlog change over the available period?
- Is customer satisfaction associated with resolution or service performance?
- Which operational areas require closer monitoring or process improvement?
- Are the available data fields consistent and reliable enough for business reporting?

4. Data Available for Analysis

Table	Purpose
company	Contains company/client details such as industry, company size, location, region, contract type and support plan.
app_user	Contains user and support-personnel information such as user type, support team, skill level, shift and location.
ticket	Main ticket table containing ticket details, dates, type, category, priority, status, source channel, reopen count and CSAT.
ticket_sla_log	Contains First Response and Resolution SLA targets, actual times, SLA status and breach minutes.
daily_itsm_kpi	Contains daily operational KPI measures such as ticket volume, backlog, SLA counts and average times.
management_itsm_summary	Contains management-level monthly measures such as ticket volume, backlog, SLA compliance, resolution time, CSAT and escalations.
Monthly activity-log tables	Contain ticket activity records for April, May, June and July 2021.

5. Expected Analytical Outcome

The analysis should provide a practical understanding of ticket demand, service performance and SLA behaviour.

The final work should not only show charts and statistics, but should also explain what the results mean from a business perspective.

The project aims to:

- Identify important workload and service-performance patterns.
- Measure relevant ITSM KPIs and trends.
- Identify areas with higher SLA risk or operational pressure.
- Highlight data-quality issues that could affect reporting.

- Convert analytical findings into practical business recommendations.

6. Scope of theProject

The project uses PostgreSQL for the database layer and Jupyter Notebook with Python for data preparation and exploratory analysis.

The required DataFrames are loaded into Jupyter and merged there using Pandas before the analysis is performed.

The analysis is organized into:

1. Data Cleaning & Preparation
2. Univariate Analysis
3. Bivariate Analysis
4. Multivariate Analysis

The final findings are documented through the EDA, Key Insights, Executive Summary and Conclusion documents.

THANKYOU!!!!
